Responsible ML System Audit

2024 HMDA Loan/Application Records | DNSC 6330

Group 4

Member	Section	Methods
Jenny	Pre-Modeling & Modeling	Data cleaning, LR, GBT, Decision Tree
Hassan	Transparency (Lecture 2)	ICE / c-ICE / d-ICE, LIME, DiCE, SHAP
Shanif	Fairness (Lecture 3)	AIR, FPR/FNR, SMD, Calibration, Intersectional
Velaphi + Jiah	Robustness (Lecture 4)	PSI, KS Test, Slice Analysis, Stress Testing
Hesham + Jiah	Privacy & Safety (Lecture 5)	PGD Evasion, Label-Flip Poisoning, Membership Inference

Executive Summary

This report presents an end-to-end Responsible Machine Learning (RML) audit of a binary classification system built on the 2024 Home Mortgage Disclosure Act (HMDA) Loan/Application Records dataset — 12,259,112 raw records covering mortgage applications across the United States. The system predicts action_taken: whether an application results in loan origination or approval (label = 1) versus denial (label = 0). After filtering to relevant outcomes, the dataset exhibits a 75.7% / 24.3% approved-to-denied split. Three model architectures were evaluated — Logistic Regression (LR), Gradient Boosted Tree (GBT), and Decision Tree (DT) — in two configurations: Model A (excluding protected attributes) and Model B (including derived_race). The selected primary model — GBT Model A — achieves AUC = 0.8533 and 84.59% test accuracy. This report addresses the five required audit questions following the Lecture 06 framework: optimization objective, failure modes, subgroup error measurement, residual risks after mitigation, and deployment defensibility.

Question 1: What is the optimization objective? (5%)

Task Definition & Label Construction

The prediction task is binary classification over HMDA mortgage applications. Labels are constructed from the action_taken field: values 1 (originated) and 2 (approved but not accepted) are collapsed into a single positive class (label = 1 — Approved), while value 3 (denied) becomes label = 0. All other categories — withdrawn, incomplete, and pre-approval requests not resulting in applications — are filtered out.

Important construct caveat: this label aggregation merges two distinct outcomes (originated vs. approved-but-not-accepted) into one positive class. That choice can hide heterogeneous applicant experience and mixes outcomes that regulators sometimes treat distinctly. It is named here as a deliberate design decision, not an oversight.

What We Optimized — Beyond a Single Number

The project frames the task as a defensible HMDA-style approval/denial decision-support exercise, not solely maximizing one global metric. ROC AUC on the held-out split is the primary development metric, as it summarizes separation between the approved and denied classes without committing to a fixed operating threshold — consistent with common underwriting analytics where posterior probability is later mapped to approvals through institutional policy.

Several fairness and robustness constructions evaluate decisions at a predicted probability threshold of 0.5. That choice is neutral for exposition: it fixes a comparator across groups and stress tests without asserting it is lender policy. Any live deployment rule would derive from institutional controls and fair-lending review, not from this notebook alone.

Model Performance Summary

Model	Train Acc.	Test Acc.	AUC	F1	Brier
LR — Model A (no race)	0.8274	0.8275	—	—	—
LR — Model B (+ race)	0.8286	0.8286	—	—	—
GBT — Model A ✓ Selected	0.8461	0.8459	0.8533	0.9051	0.1146
GBT — Model B (+ race)	0.8474	0.8473	—	—	—
Decision Tree — Model A	0.8313	0.8314	0.7902	0.8972	0.1288

GBT Model A is selected as the primary model because it achieves the highest AUC and accuracy while excluding the protected race attribute. The generalization gap is negligible (accuracy gap = 0.0001), confirming no meaningful overfitting. Including race in Model B adds only +0.001 accuracy while worsening FNR for several minority groups — providing no justification for the additional fairness cost.

Governance Implication

The objective was effectively prediction-quality-led rather than stakeholder-weighted. A defensible escalation path bundles disparity metrics and drift monitors alongside AUC with explicit escalation triggers. Absent a constrained multi-criteria objective — one that explicitly penalizes parity violations — who benefits from marginal AUC gains and who absorbs error-rate skew defaults to empirical majority structure unless actively monitored (Questions 3 through 5).

Question 2: What are the known failure modes? (10%)

Failure before deployment spans measurement, proxy, temporal validity, subgroup instability, and adversarial or pipeline manipulation. Below we name plausible modes evidenced or stress-tested in this project.

Failure Mode 1 — Label / Construct Truncation

Restricting rows to action_taken in {1, 2, 3} drops withdrawals and incomplete filings. The remaining binary approve/deny margin may not coincide with regulators' substantive questions across all workflows. Collapsing originated and approved-but-not-accepted into a single positive class further obscures heterogeneous applicant experience.

Failure Mode 2 — Proxy-Mediated Disparity Without Protected Fields

Model A omits derived race, ethnicity, and sex fields, yet geography (tract minority share), underwriting variables, and product mix correlate with demographics. ICE and SHAP analysis (Part 2) document where marginal predictions lean on surrogates. In particular, tract_minority_population_percent contributes negatively to approval probability for Black applicants (-0.20 SHAP contribution) and appears in DiCE counterfactuals as a suggested change to flip a denial — a feature the applicant cannot control.

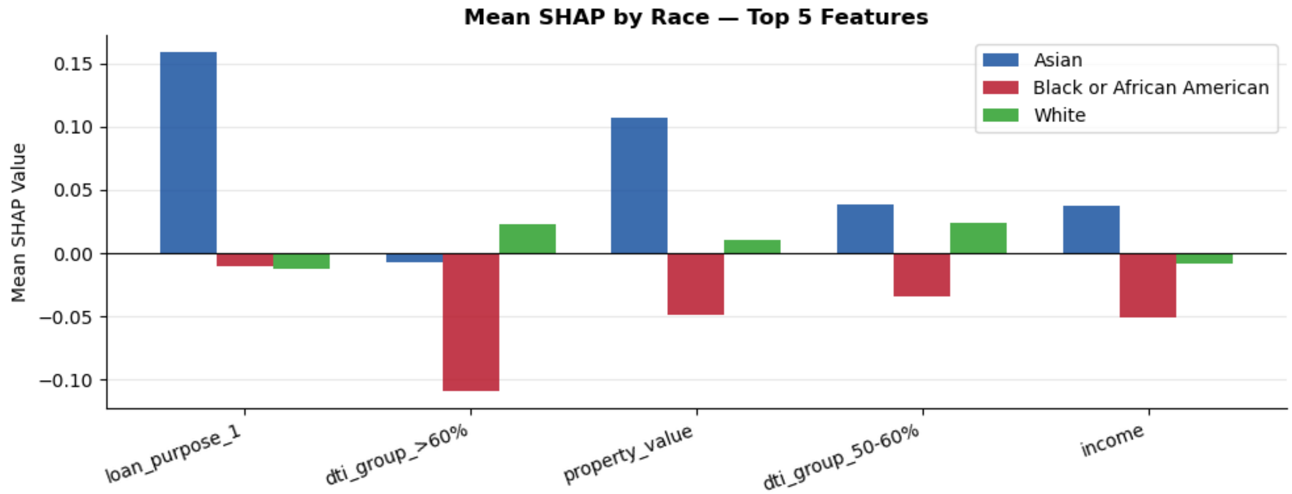

Figure 1. Mean SHAP by Race — same features produce different effect sizes across racial groups, confirming proxy-mediated disparity.

Failure Mode 3 — Subgroup Sample Sparsity

Small race or intersectional strata produce volatile FPR/FNR and unstable AIR denominators despite large overall N. Groups such as Free Form Text Only (N = 428) and 2 or more minority races (N = 4,304) have wide confidence intervals that make individual metric readings unreliable, yet they still show AIR = 0.651 and AIR = 0.884 respectively — indicating real but imprecisely estimated disparities.

Failure Mode 4 — Temporal / Cohort Drift

Part 4 uses Population Stability Index (PSI) and KS-style divergence as leading indicators when future vintages relocate in feature space versus the 2024 training cohort. While all current PSI values are 0.00 (stable), macroeconomic changes — rising interest rates, shifting property values, changing lending standards — could rapidly move these metrics and invalidate the model's learned decision boundary.

Failure Mode 5 — Adversarial Attacks Along Equity-Relevant Strata

Part 5 demonstrates two attack pathways. Under PGD-style perturbations, AIR drops from baseline 0.890 to 0.832 at $\varepsilon = 0.25$ — approaching the 4/5ths threshold — before recovering non-monotonically. Under strategic label-flip poisoning concentrated on Black or African American training records, aggregate AUC stays flat (~0.855) while AIR erodes steadily from 0.889 to 0.868 at 30% poison rate. This is the accuracy-disparity divergence path: standard AUC monitoring would detect nothing while fairness silently degrades.

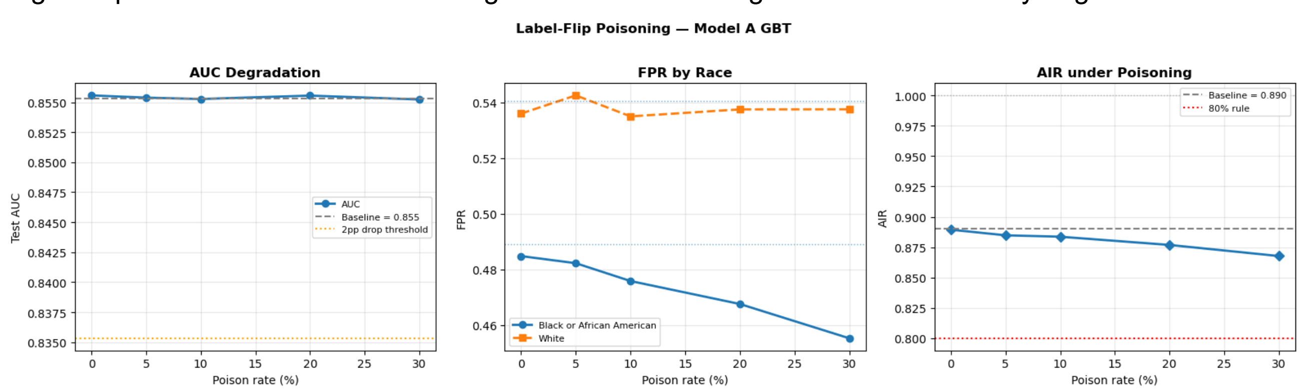

Figure 2. Label-Flip Poisoning — AUC stays flat (~0.855) while AIR steadily declines. Standard AUC monitoring would miss this entirely.

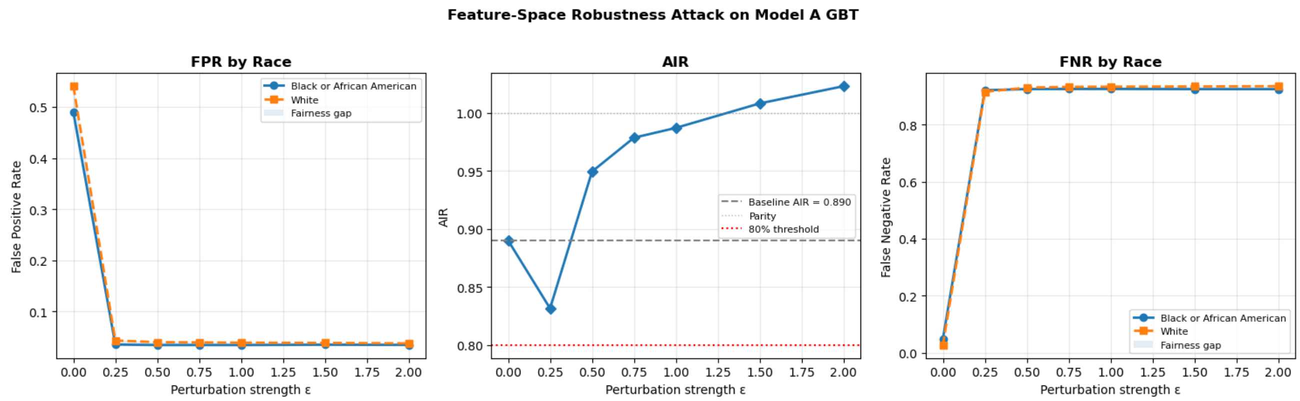

Figure 3. PGD Evasion Attack — AIR drops to 0.832 at ε = 0.25, approaching the 4/5ths threshold before recovering non-monotonically.

Failure-Mode Table (Lecture 06 Deliverable)

Type	Trigger	Affected Group(s)	Severity	Primary Evidence
Construct validity	Restrict to binary approvals vs. denial; omit other action_taken	Applicants in excluded outcome codes	Medium–High	Data prep (Part 1)
Surrogate disparity	Omit protected attributes yet retain geography / credit surrogates	Protected classes; intersectional slices	High	ICE, SHAP (Part 2); AIR/FPR/FNR (Part 3)
Subpopulation instability	Unequal N by derived_race / intersections	Rare labels (small race categories)	Medium	Subgroup tables (Part 3)
Covariate / temporal shift	Deploy period differs from training (macro, rates)	Regions or products concentrated at tails	Medium	PSI, KS (Part 4)
Robustness-fairness interaction	PGD perturbations on numeric features	Groups where score margin is thin under stress	Medium–High	PGD sweep (Part 5)
Data-pipeline poisoning	Strategic label manipulation on strata	Depends on poisoning target (UNPRIV_GROUP)	High if undetected	Poisoning sweep (Part 5)

Question 3: How are subgroup errors measured? (10%)

Fairness auditing follows the Lecture 03 framework: parity-style and classification-error viewpoints are both reported rather than flattened into a single number. Six complementary metrics are used.

1. Adverse Impact Ratio (AIR) — 4/5ths Rule

AIR compares the predicted approval prevalence of each cohort to a fixed reference group aligned with practitioner convention: White for race, Male for sex, Not Hispanic or Latino for ethnicity. Groups with AIR < 0.80 are flagged under the EEOC 4/5ths heuristic. Key results from GBT Model A: Free Form Text Only race AIR = 0.651 (⚠️ below threshold), NH/PI AIR = 0.854, Black/AA AIR = 0.892. For ethnicity: Free Form Text Only AIR = 0.783 (⚠️ below threshold). Sex: all groups pass the threshold.

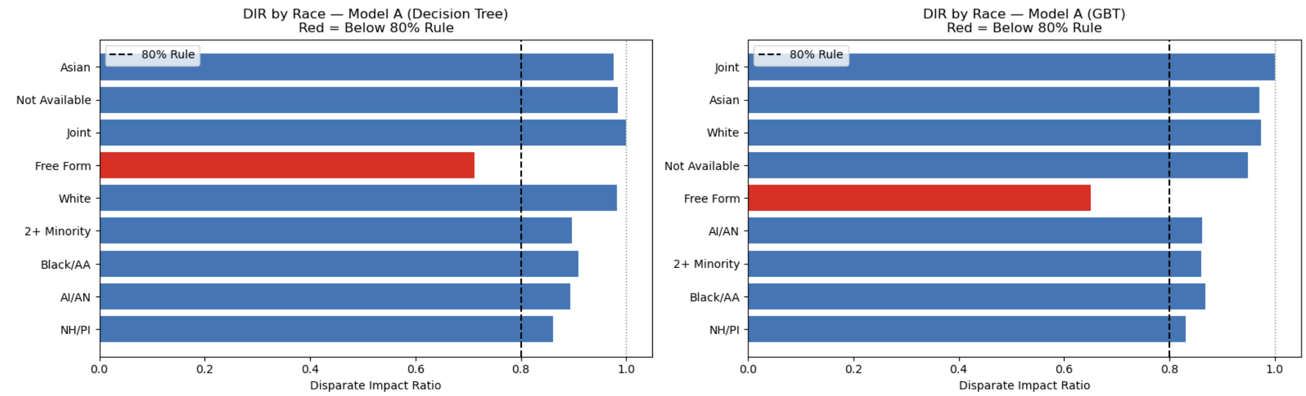

Figure 4. AIR by Race, Sex, and Ethnicity (GBT Model A). Dashed red line = 0.80 regulatory threshold.

2. Misclassification Decomposition — FPR and FNR

Subgroup FPR and FNR surface asymmetry that AUC aggregates erase. GBT Model A produces a FNR range of 0.0199–0.0526 (spread = 0.0327) across racial groups. Native Hawaiian or Other Pacific Islander exhibits the highest FNR (0.053), followed by Black or African American (0.048) — meaning qualified applicants from these groups face the highest risk of unjust denial. FPR ranges from 0.368 to 0.593. Comparisons between Decision Tree and GBT confirm these patterns are stable across architectures.

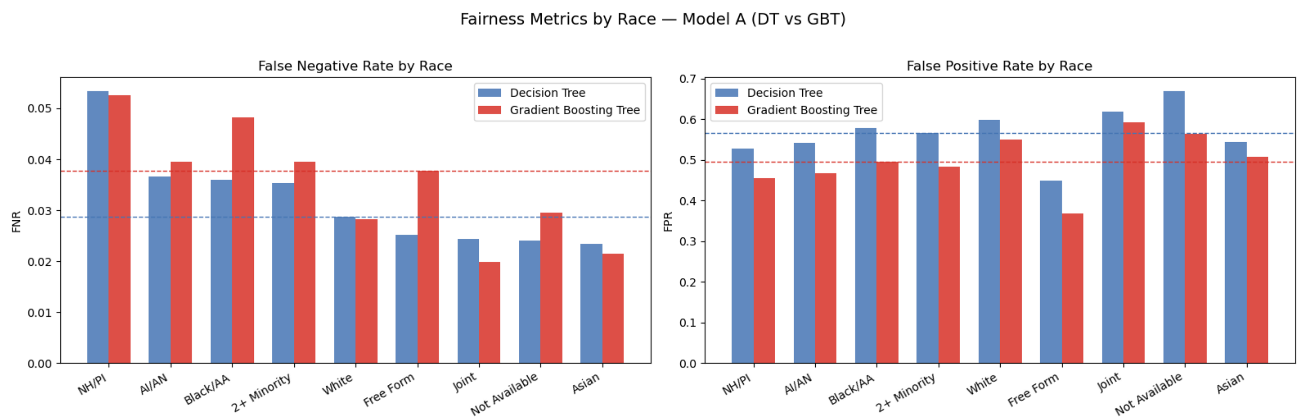

Figure 5. FNR and FPR by Race — Decision Tree vs. GBT (Model A). NH/PI and Black/AA show highest FNR under both models.

3. Standardized Mean Difference (SMD)

SMD on predicted scores versus a reference cohort quantifies separation complementary to AIR. Using White as the reference: Black or African American SMD = -0.267 (Small effect), Free Form Text Only SMD = -0.858 (Large effect). For ethnicity: Hispanic or Latino SMD = -0.174 (Negligible), Free Form Text Only SMD = -0.531 (Medium). Sex disparities are negligible across all groups (Female SMD = -0.102).

4. Calibration — Brier Score and Reliability Curves

Reliability curves check alignment between predicted and realized approval fractions; miscalibration undermines naive threshold-based remediation. White and Asian groups are closer to the perfect calibration diagonal. Black and American Indian or Alaska Native groups deviate more in the mid-probability range — meaning the model's predicted probabilities carry less reliable meaning for these groups. Even if two individuals receive the same predicted score, the actual likelihood of approval differs across groups.

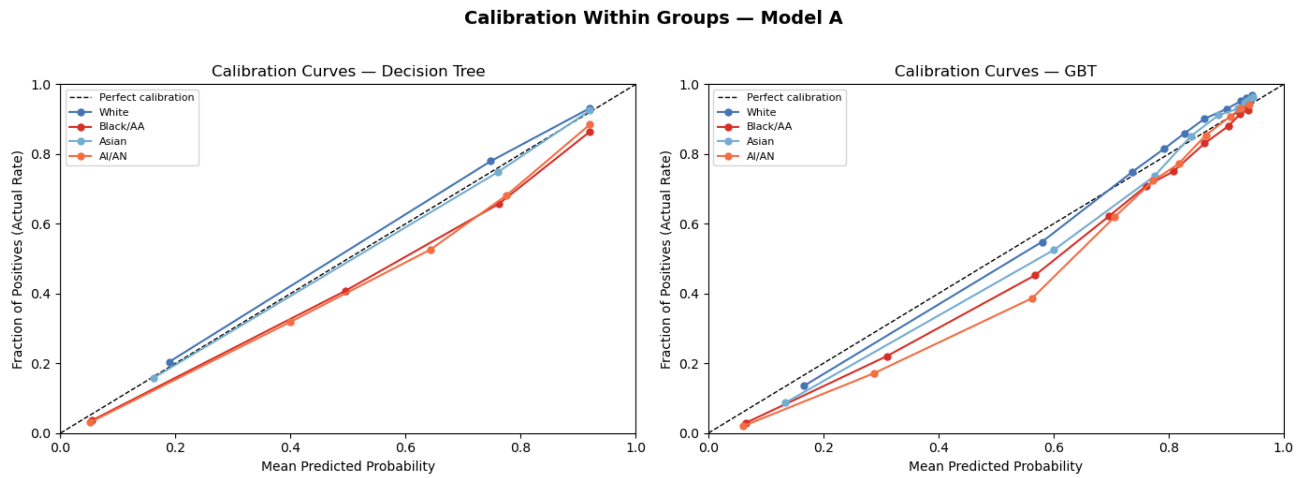

Figure 6. Brier Score by Race (DT vs GBT). Higher Brier = less-calibrated predictions. Minority groups show higher values.

5. SHAP Group-Level Feature Contributions

Mean SHAP values disaggregated by race reveal that the same features produce different effect sizes across groups. DTI > 60% has a strongly negative effect for Black applicants but near-zero or positive for White applicants. Property value contributes positively for Asian and White applicants but negatively for Black applicants. These group-level differences are direct evidence of heterogeneous model treatment through proxy channels.

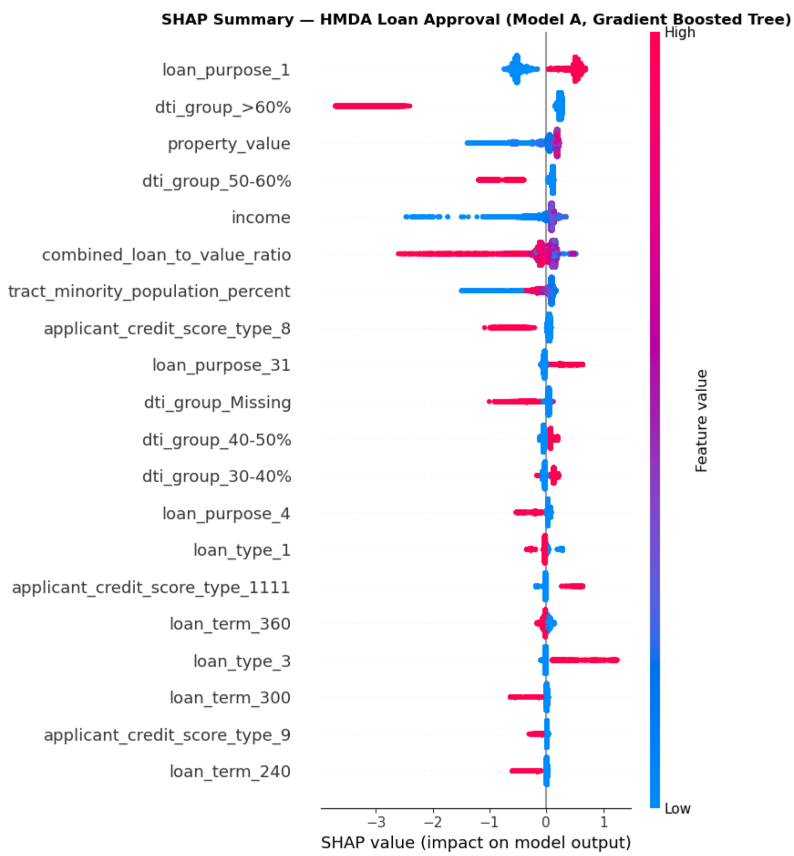

Figure 7. SHAP Beeswarm Summary — DTI, income, loan purpose, and property value are the top drivers.

6. Intersectional Views — Race × Sex

Joint race-by-sex intersections mitigate narrow non-intersectional averages that can mask harm on smaller cells. Native Hawaiian or Other Pacific Islander / Female and Black or African American / Female exhibit the highest FNR values across both models — compounded disadvantage that is invisible in single-axis analysis. White Male and Asian Male achieve the highest predicted approval rates.

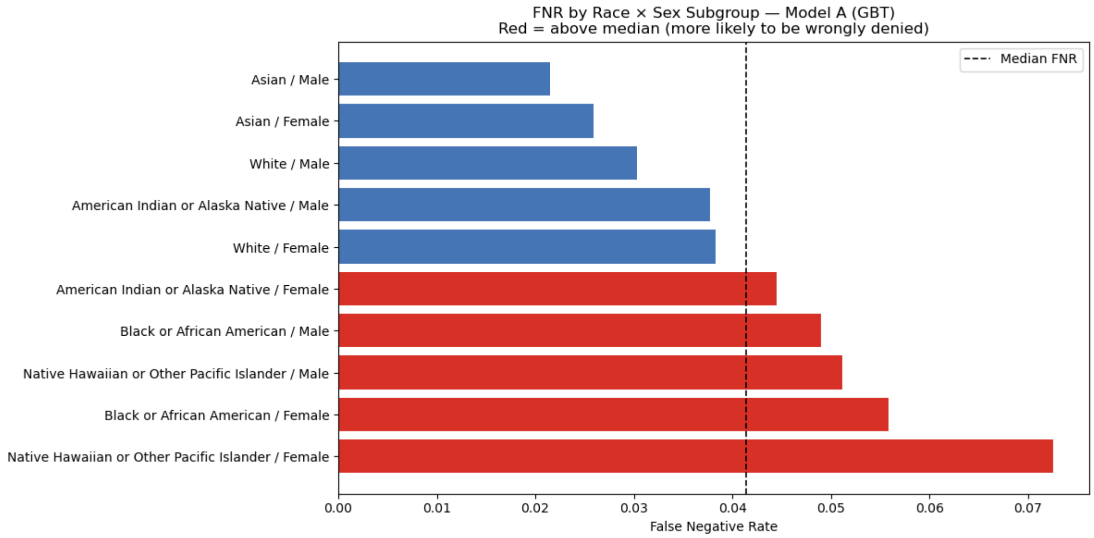

Figure 8. Intersectional Fairness (Race × Sex) — FNR comparison. Black/AA Female and NH/PI Female show the highest combined FNR.

Distribution Drift Validation: PSI & KS

PSI and KS tests confirm no significant distribution shift between training and test sets. All features show PSI = 0.00 (well below the 0.10 warning threshold). All KS p-values exceed 0.05. These results confirm that observed subgroup disparities reflect genuine model behavior, not train-test distributional artifacts.

Question 4: What risks remain after mitigation? (3%)

Mitigations evidenced in this project include: omitting protected fields in Model A, SHAP-guided transparency for proxy detection, disparity tables with statistical tests, PSI/KS monitoring, adversarial probing (Parts 4–5), and Lecture 06 references to reweighting, threshold adjustment, and differential privacy caveats.

Acceptance principle: only small residual disparities are acceptable, and then only if they are documented, monitored, and justified as an explicit trade-off — not treated as noise. Larger or widening gaps must drive remediation or pause.

Residual Risk 1 — Proxy Disparity Persists

Despite race omission, AIR for Black/AA = 0.892 and FNR for NH/PI reaches 0.053. The TMPP (tract minority population percentage) proxy effect is systematic and monotonic: approval rates decrease as TMPP increases from Q1 to Q4, with FNR highest in high-TMPP areas. Removing protected attributes is insufficient — proxy channels remain active through correlated financial and geographic features.

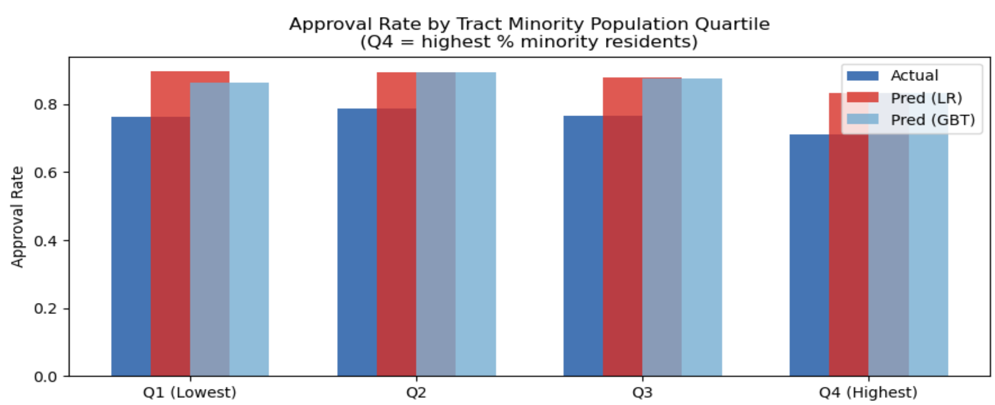

Figure 9. Approval rate by Tract Minority Population Quartile. Q4 (highest minority %) shows lowest approval — systematic proxy effect.

Residual Risk 2 — Monitoring Ownership Is Procedural

PSI and KS monitoring concepts are evidenced in Part 4, but monitoring ownership is implied, not enforced in the repository. There is no automated alert system, no named owner for escalation, and no defined SLA for rollback. This is a governance gap, not a technical gap.

Residual Risk 3 — Poisoning Stealth Zone

Part 5 demonstrates that AUC stays near 0.855 across all tested poison rates while AIR declines from 0.889 to 0.868 at 30% poison rate. A production system monitoring only AUC would observe no anomaly throughout this degradation. This is the stealth zone: fairness deteriorates while performance appears stable. This risk is not acceptable absent dual monitoring of disparity indices alongside AUC.

Residual Risk 4 — CLTV Stress Sensitivity

Under CLTV +10 percentage points stress, AUC drops from 0.8533 to 0.8196 and FNR increases from 0.030 to 0.050. Income and property value shocks show minimal impact. The minimum group approval rate under CLTV stress falls to 0.575, disproportionately affecting groups with higher average leverage.

Residual Risk 5 — Privacy (Configuration-Dependent)

Membership inference achieved MI AUC ≈ 0.50 — equivalent to random guessing — under the current configuration (max_depth = 3, n_estimators = 100). The near-zero generalization gap (0.0004) explains the null signal. However, this result is configuration-dependent: deeper trees or larger ensembles could reopen the privacy attack surface unless explicitly regularized.

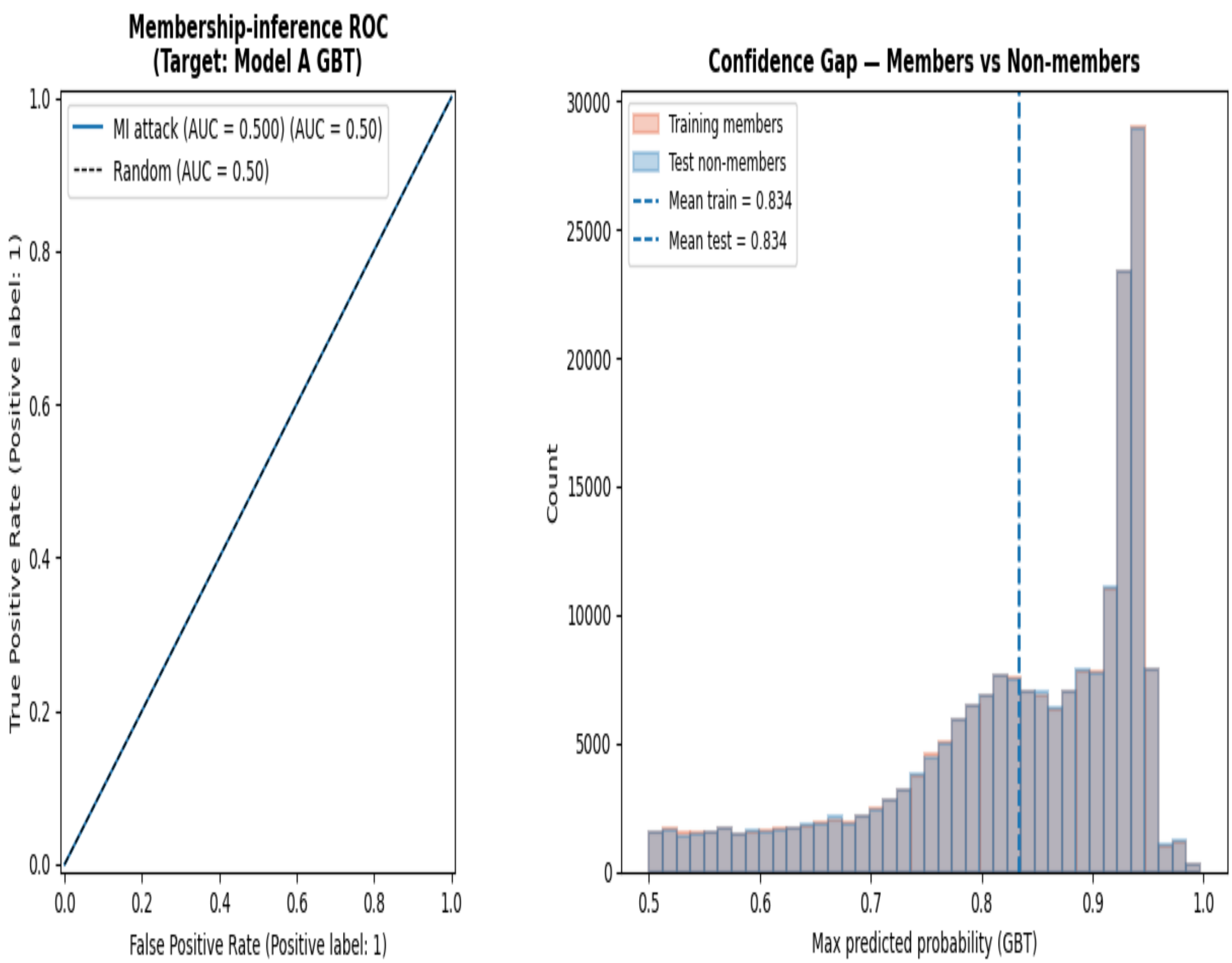

Figure 10. Membership Inference — confidence distributions for training members vs. non-members overlap completely. MI AUC ≈ 0.50.

Residual Risk Register (Lecture 06 Deliverable)

Risk	Mitigation Attempted	Observed Outcome	Accepted Residual
Proxy disparity without protected attributes	Model A omission; SHAP; proxy analysis	Gaps persist in AIR, FPR, FNR across race groups	Accepted only under live disparity triggers; escalate if widening
Distribution shift vs. 2024 training	PSI, KS trackers (Part 4)	All PSI = 0.00; all KS p > 0.05 currently	Accepted with named alert owner; rollback on PSI > 0.25 or SLA breach
Stealth disparity under poisoning	Dual monitoring of AIR alongside AUC (Part 5)	Accuracy-disparity divergence confirmed at 30% poison	Not acceptable absent dual-metric production monitoring
CLTV stress sensitivity	Scenario stress testing (Part 4)	AUC drops to 0.82 at CLTV +10pp	Accepted with defined CLTV drift trigger (>5pp market shift → review)
Privacy / memorization	MI on mirrored GBT; generalization gap monitored	MI AUC ≈ 0.50; gap = 0.0004 (negligible)	Accepted under current depth=3 config; revisit on architecture changes

Question 5: Is deployment defensible? (2%)

Overall Fairness Summary

The fairness analysis covering all required metrics yields the following key findings. On equalized odds: racial disparities persist across both DT and GBT — DT achieves lower FNR while GBT achieves lower FPR, confirming that model choice affects the type of error different groups experience rather than eliminating disparities. On AIR: Free Form Text Only groups fall below the 4/5ths threshold for both race and ethnicity. On calibration: White and Asian groups receive better-calibrated probability estimates than minority groups. On intersectionality: Black/AA and NH/PI female applicants face compounded FNR penalties invisible in single-axis analysis.

Robustness Confirmation

Both models exhibit negligible train-test gaps (GBT: accuracy gap = 0.0001, AUC gap = -0.0001), confirming no meaningful overfitting. PSI = 0.00 across all numeric features. All KS p-values > 0.05. The model's performance and fairness evaluations are not driven by distribution drift — observed disparities reflect genuine model behavior.

Deployment Verdict: Pilot-Ready, Not Fully Autonomous

Outputs from this audit support a pilot-ready measurement package with explicit limits. A production-facing path would resemble a conditional approval or a controlled pilot: model outputs inform a scorecard-style or human-in-the-loop process, not a silent, ungated auto-decision rule by default. The model should not operate as a standalone approval engine without human oversight at the decision boundary.

Pause and rollback triggers must be defined before any deployment. Specifically:

- AIR or subgroup FPR/FNR beyond agreed tolerances vs. baseline (e.g., AIR dropping below 0.85 triggers review; below 0.80 triggers suspension)
- PSI/KS drift breaches (PSI > 0.10 triggers monitoring; PSI > 0.25 triggers mandatory retraining)
- CLTV stress sensitivity (market CLTV shift > 5pp triggers model review)
- Security findings from Part 5 (AIR erosion detected in poisoning sweep → activate dual-metric monitoring immediately)
- AUC alone is insufficient as a production monitor — the poisoning experiment confirms fairness can degrade while AUC stays flat

Institutional practice would add ongoing monitoring dashboards, periodic re-validation with challenger models, and the option to de-model or restrict model use in segments where monitors breach policy. Statutory sign-off remains outside this classroom replication and requires lender-specific legal and compliance review.

Required Governance Guardrails

Guardrail	Evidence from Audit	Trigger
Monthly AIR monitoring by race/sex/ethnicity	Poisoning stealth: AIR erodes while AUC = 0.855	AIR < 0.85 → review; < 0.80 → suspend
FNR parity dashboard	NH/PI FNR = 0.053, Black/AA FNR = 0.048	FNR spread > 0.04 → audit
PSI input monitoring (quarterly)	Baseline PSI = 0.00; all KS p > 0.05	PSI > 0.10 → warning; > 0.25 → retrain
CLTV stress watch	CLTV +10pp drops AUC from 0.853 to 0.820	Market CLTV shift > 5pp → review
Dual-metric production monitor	AUC alone cannot detect poisoning	Track AIR + AUC together in production
ECOA adverse action notices	Regulation B compliance required	All denial decisions — SHAP supports explanation
Human-in-the-loop at decision boundary	Model outputs are probability scores, not policy	No ungated auto-decision rule without review

Suggested Presentation Order — Lecture 06 Audit Sequence

When presenting this audit to a regulator or reviewer, the recommended order is:

- (1) Label and objective transparency — establish what was optimized and what trade-offs were made
- (2) AIR / FPR / FNR / intersections — lead with the disparity evidence
- (3) SHAP proxy story — explain how surrogates carry demographic signal despite race exclusion
- (4) Drift (PSI / KS) — confirm the model is being evaluated on a representative population
- (5) Security stress (PGD, poisoning, MI) — demonstrate adversarial awareness
- (6) Residual risk register and escalation — show governance is in place
- (7) Go / Pilot / No-go — framed as fair-lending and safety criteria, not AUC-only

Course Recommendation

Outputs support a pilot-ready measurement package with explicit limits. A production-facing path would resemble a conditional approval or a controlled pilot: model outputs inform a scorecard-style or human-in-the-loop process, not a silent, ungated auto-decision rule by default. Pause and rollback triggers should be defined up front — for example: AIR or subgroup FPR/FNR beyond agreed tolerances versus baseline, PSI/KS drift breaches, material Part 5 security findings, or compliance flags. Institutional practice would add ongoing monitoring, periodic re-validation or challenger models, and the option to de-model or restrict model use in segments where monitors breach policy. Statutory sign-off remains outside this classroom replication and requires lender-specific legal and compliance review.

Regulatory Alignment

- Equal Credit Opportunity Act (ECOA / Regulation B) — non-discrimination in credit decisions; written adverse action notices required for all denials
- Fair Housing Act (FHA) — prohibition of discriminatory practices in residential mortgage lending
- NIST AI Risk Management Framework (AI RMF) — GOVERN, MAP, MEASURE, MANAGE pillars
- NIST AI 100-2 — adversarial ML threat taxonomy: evasion (PGD), poisoning (label-flip), inference (membership)
- EEOC 4/5ths Rule (UGESP) — AIR ≥ 0.80 required; Free Form Text group currently below threshold for race and ethnicity